



Structural Equation Modeling: A Review of Its History, Methods, and Applications

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ABSTRACT

Background: Structural equation modeling (SEM) is a family of multivariate statistical techniques that allows researchers to specify, estimate, and test networks of relationships among observed and latent variables. Combining the measurement logic of factor analysis with the causal-relational logic of path analysis, SEM has become one of the dominant analytic frameworks in the social, behavioral, health, and management sciences, yet its historical development, statistical foundations, and rapidly evolving software ecosystem are rarely synthesized in one place.

Method: This paper presents a narrative review tracing the historical development of SEM from Sewall Wright's early path-analytic work in genetics, through the psychometric contributions of Spearman and Thurstone, to the unification achieved by Jöreskog, Keesling, and Wiley and the subsequent diffusion of the LISREL program. It surveys the core statistical building blocks of SEM, including model specification and identification, sample size and power considerations, estimation methods (maximum likelihood, weighted least squares, and Bayesian estimation), and the fit indices commonly used to evaluate model adequacy, alongside the major SEM variants and the software ecosystem that supports them.

Result: The review synthesizes SEM's major methodological variants – confirmatory factor analysis, path analysis, multi-group and longitudinal SEM, mediation and moderation models, partial least squares SEM, Bayesian SEM, exploratory structural equation modeling, and network psychometric alternatives – together with applications across psychology, education, business, and health research. It further documents recent developments, including dynamic structural equation modeling for intensive longitudinal data, hybrid SEM–machine learning approaches, and the expanding open-source R ecosystem (*lavaan*, *SEMinR*, *cSEM*, *blavaan*).

Conclusion: SEM remains an active and evolving methodological field rather than a fixed, uncontested technique. Researchers using SEM are best served by grounding their modeling choices in both their substantive theory and a clear understanding of the assumptions and trade-offs associated with the particular SEM variant, estimator, and software tool they adopt.

Key words: structural equation modeling, path analysis, confirmatory factor analysis, latent variables, PLS-SEM, Bayesian SEM, model fit, sample size

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ABSTRACT

Structural equation modeling (SEM) is a family of multivariate statistical techniques that allows researchers to specify, estimate, and test networks of relationships among observed and latent variables. Combining the measurement logic of factor analysis with the causal-relational logic of path analysis, SEM has become one of the dominant analytic frameworks in the social, behavioral, health, and management sciences. This review traces the historical development of SEM from Sewall Wright's early path-analytic work in genetics (Wright, 1918, 1934), through the psychometric contributions of Spearman and Thurstone, to the unification achieved by Jöreskog, Keesling, and Wiley (Jöreskog, 1969, 1973) and the subsequent diffusion of the LISREL program (Bollen, 1989; Jöreskog and Sörbom, 1978). It then surveys the core statistical building blocks of SEM, including model specification and identification, sample size and power considerations, estimation methods such as max-

imum likelihood, weighted least squares, and Bayesian estimation, and the fit indices commonly used to evaluate model adequacy (Hu and Bentler, 1999; Schermelleh-Engel et al., 2003; Steiger, 2007). The review further examines major SEM variants, including confirmatory factor analysis, path analysis, multi-group and longitudinal SEM, mediation and moderation models, partial least squares SEM (Hair et al., 2011; Wold, 1982), Bayesian SEM (Muthén and Asparouhov, 2012), exploratory structural equation modeling (Marsh et al., 2009), and network psychometric alternatives (Borsboom et al., 2003; Epskamp et al., 2018), alongside the software ecosystem that supports them, including the open-source *lavaan*, *SEMinR*, *cSEM*, and *blavaan* packages (Danks and Ray, 2020; Merkle and Rosseel, 2018; Rademaker and Schuberth, 2020; Rosseel, 2012). Applications across psychology, education, business, and health research are summarized (MacCallum and Austin, 2000), followed by a discussion of the strengths, limitations, and controversies

surrounding the technique. The review closes with recent developments – dynamic structural equation modeling for intensive longitudinal data (Asparouhov et al., 2018), hybrid SEM–machine learning approaches (Grassi et al., 2022; Richter and Tudoran, 2024), and multimethod CB-SEM/PLS-SEM practice (Guenther et al., 2025) – and an outlook on emerging directions.

Keywords: structural equation modeling, path analysis, confirmatory factor analysis, latent variables, PLS-SEM, Bayesian SEM, model fit, sample size

INTRODUCTION

Structural equation modeling (SEM) refers to a broad class of statistical techniques used to test hypothesized relationships among multiple variables simultaneously, including relationships that involve unobserved, or latent, constructs. Unlike traditional regression, which examines one dependent variable at a time and generally assumes that predictors are measured without error, SEM allows researchers to model measurement error explicitly, to estimate multiple dependent relationships in a single analysis, and to compare an implied theoretical structure against the pattern of covariances actually observed in the data. Because of this flexibility, SEM has become a standard tool wherever researchers work with abstract constructs – attitudes, personality traits, organizational climate, quality of life, or economic expectations – that cannot be measured directly but can be inferred from multiple observed indicators (MacCallum and Austin, 2000).

The appeal of SEM lies in its integrative character: it merges the measurement tradition of factor analysis, which is concerned with how well observed items represent an underlying construct, with the structural tradition of path analysis and simultaneous-equation modeling, which is concerned with the pattern of causal or predictive relationships among variables. This review synthesizes the historical origins of SEM, its statistical foundations, its major methodological variants, the software that implements it, its applications across disciplines, and the debates that continue to shape its use.

HISTORICAL DEVELOPMENT OF STRUCTURAL EQUATION MODELING

Early Roots: Path Analysis and Factor Analysis

The statistical lineage of SEM can be traced to two largely independent traditions that developed in the early twentieth century. The first is path analysis, introduced by the geneticist Sewall Wright, whose 1918 study of skeletal proportions in rabbits used path diagrams to represent hypothesized causal relationships among variables and compared the correlations implied by those diagrams with the correlations actually observed (Wright, 1918). Wright's method, elaborated further in his work on the method of path coefficients (Wright, 1934), provided a way to decompose an observed correlation into direct and indirect components attributable to a specified causal structure, anticipating the decomposition of effects into direct, indirect, and total effects that remains

central to SEM today.

The second tradition is factor analysis, originating with Charles Spearman's early twentieth-century work on general intelligence and extended in the 1930s by Louis Thurstone's multiple-factor model and his development of factor rotation. Where path analysis modeled relationships among directly measured variables, factor analysis modeled the extent to which a smaller number of unobserved factors could account for the pattern of correlations among a larger set of observed indicators.

Although Wright's path-analytic approach attracted little attention among psychologists for several decades, it was adopted more readily by sociologists and economists. Sociologists such as Hubert Blalock (Blalock, 1964) and Otis Dudley Duncan (Duncan, 1966) popularized path analysis within the social sciences during the 1960s, while economists working within the Cowles Commission tradition, building on Wright's ideas, developed related simultaneous-equation approaches for econometric modeling.

Unification: Jöreskog, Keesling, Wiley, and LISREL

The modern, unified form of SEM emerged in the late 1960s and early 1970s through the work of the psychometrician Karl Jöreskog, who introduced confirmatory factor analysis (CFA) with maximum-likelihood estimation (Jöreskog, 1969), enabling formal statistical tests of hypotheses about the number of underlying factors and how they relate to observed variables. Jöreskog, together with contributions from Keesling and Wiley in the early 1970s, generalized this framework to combine the measurement structure of factor analysis with the structural, path-analytic modeling of relationships among latent variables (Jöreskog, 1973), producing what became known as the Jöreskog–Keesling–Wiley model – the general structural equation model that underlies contemporary SEM.

This theoretical unification was accompanied by a practical breakthrough: the development of the LISREL (Linear Structural RELations) computer program by Jöreskog and Dag Sörbom (Jöreskog and Sörbom, 1978). LISREL gave applied researchers, for the first time, an accessible and efficient way to specify and estimate structural equation models. Bollen (1989) argued that the availability of LISREL was itself one of the most important factors in the rapid spread of SEM techniques through the social and behavioral sciences during the 1970s and 1980s, since it allowed researchers without specialized mathematical training to fit and evaluate complex latent variable models.

Diffusion and Diversification

From the 1980s onward, SEM diffused well beyond its psychometric and sociological origins into education, management, marketing, political science, epidemiology, and health services research (MacCallum and Austin, 2000). This diffusion was accompanied by methodological diversification: new estimators were developed to handle non-normal and categorical data, new fit indices were proposed and evaluated (Hu and Bentler, 1999), and alternative estimation traditions – most notably partial least squares path modeling, introduced by the econometrician

and statistician Herman Wold as a composite-based alternative to Jöreskog's factor-based approach (Wold, 1982) – expanded the range of situations in which SEM-type analysis could be applied. In more recent decades the field has continued to generalize further, incorporating multilevel structures, longitudinal growth models, mixture models, Bayesian estimation (Muthén and Asparouhov, 2012), and, increasingly, connections to network-based approaches to modeling psychological constructs (Ep-skamp et al., 2018).

THEORETICAL AND STATISTICAL FOUNDATIONS

The Measurement Model and the Structural Model

A complete structural equation model is generally understood as consisting of two interlocking components. The *measurement model* specifies how each latent variable is indicated by one or more observed variables, typically through a confirmatory factor-analytic structure that estimates factor loadings and accounts for measurement error in each indicator. The *structural model* specifies the hypothesized directional and non-directional relationships among the latent variables themselves, analogous to a system of regression equations but operating on error-free latent constructs rather than raw observed scores. This separation of concerns is one of the principal advantages of SEM over ordinary regression: because measurement error is modeled explicitly in the measurement component, the structural relationships among constructs can, in principle, be estimated without attenuation bias due to unreliable measurement (Anderson and Gerbing, 1988).

Following the classic LISREL notation introduced by Jöreskog (1973), the measurement model relates a vector of observed indicators $\mathbf{x}$ to a vector of exogenous latent variables $\boldsymbol{\xi}$, and a second vector of observed indicators $\mathbf{y}$ to a vector of endogenous latent variables $\boldsymbol{\eta}$:

$$\mathbf{x} = \mathbf{\Lambda}_x \boldsymbol{\xi} + \boldsymbol{\delta}, \quad (1)$$

$$\mathbf{y} = \mathbf{\Lambda}_y \boldsymbol{\eta} + \boldsymbol{\varepsilon}, \quad (2)$$

where $\mathbf{\Lambda}_x$ and $\mathbf{\Lambda}_y$ are matrices of factor loadings, and $\boldsymbol{\delta}$ and $\boldsymbol{\varepsilon}$ are vectors of measurement errors assumed to have zero mean and to be uncorrelated with the latent variables, i.e. $E(\boldsymbol{\delta}) = \mathbf{0}$, $E(\boldsymbol{\varepsilon}) = \mathbf{0}$, $\text{Cov}(\boldsymbol{\xi}, \boldsymbol{\delta}) = \mathbf{0}$, and $\text{Cov}(\boldsymbol{\eta}, \boldsymbol{\varepsilon}) = \mathbf{0}$.

The structural model then specifies the hypothesized causal relationships among the latent variables themselves:

$$\boldsymbol{\eta} = \mathbf{B}\boldsymbol{\eta} + \mathbf{\Gamma}\boldsymbol{\xi} + \boldsymbol{\zeta}, \quad (3)$$

where $\mathbf{B}$ contains structural coefficients among endogenous latent variables (with zeros on the diagonal, and $(\mathbf{I} - \mathbf{B})$ assumed invertible), $\mathbf{\Gamma}$ contains structural coefficients relating exogenous to endogenous latent variables, and $\boldsymbol{\zeta}$ is a vector of disturbances (structural errors) uncorrelated with $\boldsymbol{\xi}$.

Under this specification, the model-implied population covariance matrix of the observed variables, $\boldsymbol{\Sigma}(\boldsymbol{\theta})$, is a function of the free parameters $\boldsymbol{\theta} = \{\mathbf{\Lambda}_x, \mathbf{\Lambda}_y, \mathbf{B}, \mathbf{\Gamma}, \boldsymbol{\Phi}, \boldsymbol{\Psi}, \boldsymbol{\Theta}_\delta, \boldsymbol{\Theta}_\varepsilon\}$, where $\boldsymbol{\Phi} = \text{Cov}(\boldsymbol{\xi})$, $\boldsymbol{\Psi} =$

$\text{Cov}(\boldsymbol{\zeta})$, and $\boldsymbol{\Theta}_\delta$, $\boldsymbol{\Theta}_\varepsilon$ are the (co)variance matrices of the measurement errors. For the endogenous block, substituting Equation (3) and solving for $\boldsymbol{\eta}$ gives $\boldsymbol{\eta} = (\mathbf{I} - \mathbf{B})^{-1}(\mathbf{\Gamma}\boldsymbol{\xi} + \boldsymbol{\zeta})$, from which the implied covariance structure follows as

$$\boldsymbol{\Sigma}(\boldsymbol{\theta}) = \begin{pmatrix} \mathbf{\Lambda}_y \mathbf{A}(\mathbf{\Gamma}\boldsymbol{\Phi}\mathbf{\Gamma}' + \boldsymbol{\Psi})\mathbf{A}'\mathbf{\Lambda}_y' + \boldsymbol{\Theta}_\varepsilon & \mathbf{\Lambda}_y \mathbf{A}\mathbf{\Gamma}\boldsymbol{\Phi}\mathbf{\Lambda}_x' \\ \mathbf{\Lambda}_x \boldsymbol{\Phi}\mathbf{\Gamma}'\mathbf{A}'\mathbf{\Lambda}_y' & \mathbf{\Lambda}_x \boldsymbol{\Phi}\mathbf{\Lambda}_x' + \boldsymbol{\Theta}_\delta \end{pmatrix}, \quad (4)$$

Estimation, described in Section 3.4, seeks values of $\boldsymbol{\theta}$ that make $\boldsymbol{\Sigma}(\boldsymbol{\theta})$ as close as possible to the sample covariance matrix $\mathbf{S}$.

Model Specification and Identification

Before a structural equation model can be estimated, the researcher must specify which parameters (factor loadings, structural paths, variances, covariances) are to be freely estimated, which are fixed to particular values, and which are constrained to be equal to one another. This specification must yield a model that is *identified* – that is, one for which a unique set of parameter estimates can, in principle, be recovered from the observed covariance matrix. A necessary (though not sufficient) condition, commonly called the *t-rule*, requires that the number of freely estimated parameters, t , not exceed the number of non-redundant elements in the observed covariance matrix. For p observed variables there are

$$s = \frac{p(p+1)}{2} \quad (5)$$

distinct variances and covariances, so the model degrees of freedom are

$$df = s - t = \frac{p(p+1)}{2} - t. \quad (6)$$

A model with $df \geq 0$ is said to be *over-identified* (or *just-identified* when $df = 0$); a model with $df < 0$ is under-identified and cannot be estimated without further constraints. In addition to satisfying the *t-rule*, each latent variable must be assigned a metric, typically by fixing one factor loading to 1 (the *marker-variable* method) or by standardizing the latent variable's variance to 1, since latent variables have no inherent scale of their own.

Sample Size and Statistical Power

Because SEM typically estimates many parameters simultaneously and relies on asymptotic (large-sample) theory for maximum-likelihood inference, adequate sample size is a recurring practical concern. Applied researchers have long relied on simple rules of thumb: an absolute minimum of roughly 100 to 200 cases (Kline, 2015), or a ratio of cases to freely estimated parameters (the “N:q rule”), with recommendations for this ratio ranging from as low as 5:1 to as high as 20:1 depending on the source (Jackson, 2003; Kline, 2015). Such rules, however, ignore model-specific features that strongly influence the precision of estimates and the power of fit tests, including the number of indicators per factor, the size of factor loadings, and the degree of model misspecification being tested against.

More principled approaches instead conduct formal power analysis. MacCallum et al. (1996) developed

a widely used framework for power analysis based on the root-mean-square error of approximation (RMSEA), which tests hypotheses of close fit (e.g., $\varepsilon_0 \leq 0.05$) against not-close fit (e.g., $\varepsilon_1 \geq 0.08$) as a function of sample size, degrees of freedom, and significance level. The procedure relies on the fact that, under a fixed alternative RMSEA value ε_a , the likelihood-ratio test statistic follows a non-central chi-square distribution with noncentrality parameter

$$\lambda = (N - 1) df \varepsilon_a^2, \quad (7)$$

where N is the sample size and df the model degrees of freedom. Power for the test of close fit is then obtained as $1 - \beta = P(\chi_{df, \lambda}^2 > \chi_{df, 1-\alpha}^2)$, and the researcher can solve for the minimum N that achieves a target power (typically 0.80) for a given df , ε_0 , ε_1 , and α .

The simpler, more commonly reported N:q heuristic instead sets the required sample size as a fixed multiple k of the number of freely estimated parameters t :

$$N_{\min} = k \cdot t, \quad k \in \{5, 10, 20\}, \quad (8)$$

with $k = 20$ recommended by Kline (2015), $k = 10$ by other sources, and k as low as 5 suggested in some older guidelines; Jackson (2003) showed via simulation that the appropriate value of k itself depends on communality and the number of indicators per factor, so a single fixed ratio is, at best, an approximation. More recently, Wang and Rhemtulla (2021) argued that researchers should shift emphasis from power to detect overall model misfit toward power to detect specific target parameters of substantive interest, such as an individual structural path or indirect effect γ_{ij} , using the Wald-test-based approximation

$$1 - \beta \approx P\left(Z > z_{1-\alpha/2} - \frac{\gamma_{ij}}{SE(\hat{\gamma}_{ij})}\right), \quad (9)$$

and provided simulation-based tools for this purpose. Together, this body of work suggests that a single universal sample-size rule cannot substitute for power analysis tailored to a given model and research question.

Estimation Methods

Maximum likelihood (ML) estimation is the default and most widely used estimator in SEM. Under the assumption that the p observed variables follow a multivariate normal distribution, ML estimation chooses the parameter vector $\hat{\theta}$ that minimizes the discrepancy function

$$F_{ML}(\theta) = \log |\Sigma(\theta)| + \text{tr}[\mathbf{S} \Sigma(\theta)^{-1}] - \log |\mathbf{S}| - p, \quad (10)$$

where $\mathbf{S}$ is the sample covariance matrix and $\Sigma(\theta)$ is the model-implied covariance matrix from Equation (4). At the minimum, $F_{ML}(\hat{\theta}) \geq 0$, with equality only when the model reproduces $\mathbf{S}$ exactly. The overall chi-square test statistic used to evaluate exact fit is then

$$\chi^2 = (N - 1) F_{ML}(\hat{\theta}), \quad (11)$$

which, under the null hypothesis that the model holds exactly in the population and standard regularity conditions, is asymptotically distributed as χ_{df}^2 with df as defined above. When the multivariate normality assumption is violated, robust variants such as MLM and MLR

rescale this statistic (the Satorra–Bentler correction) and adjust standard errors to account for excess kurtosis. For categorical or ordinal indicators, estimators from the weighted least squares family, particularly the diagonally weighted least squares estimator with robust corrections (often labeled WLSMV), minimize a discrepancy function of the form $F_{WLS} = (\mathbf{s} - \sigma(\theta))' \mathbf{W}^{-1} (\mathbf{s} - \sigma(\theta))$, where $\mathbf{s}$ and $\sigma(\theta)$ are vectorized sample and model-implied polychoric/polyserial correlations and $\mathbf{W}$ is a weight matrix; these are commonly preferred over ML for ordinal data because they do not require treating ordered-categorical responses as continuous.

An increasingly prominent alternative is Bayesian estimation, popularized within the SEM literature by Muthén and Asparouhov (2012)'s proposal to replace exact zero-constraints – for example, cross-loadings or residual correlations fixed at zero for identification purposes – with small-variance informative priors that allow such parameters to be approximately, rather than exactly, zero (Asparouhov et al., 2015). Rather than maximizing the likelihood $L(\mathbf{X} | \theta)$ of the raw data $\mathbf{X}$ directly, Bayesian SEM combines it with a prior distribution $p(\theta)$ over the model parameters, via Bayes' theorem, to obtain the posterior distribution

$$p(\theta | \mathbf{X}) = \frac{L(\mathbf{X} | \theta) p(\theta)}{\int L(\mathbf{X} | \theta) p(\theta) d\theta}, \quad (12)$$

typically approximated via Markov chain Monte Carlo (MCMC) sampling rather than solved in closed form. For an approximate zero-constraint, a parameter ϕ (e.g., a cross-loading) that would be fixed at exactly 0 under ML is instead assigned a small-variance normal prior, $\phi \sim N(0, \sigma_0^2)$ with σ_0^2 chosen to be small (e.g., $\sigma_0^2 = 0.01$), so that the posterior for ϕ is shrunk toward, but not forced to equal, zero. Point estimates are then commonly summarized by the posterior mean or median, and uncertainty by the 95% credibility interval, i.e. the interval $[\theta_{2.5\%}, \theta_{97.5\%}]$ such that $P(\theta_{2.5\%} \leq \phi \leq \theta_{97.5\%} | \mathbf{X}) = 0.95$. Advocates note that Bayesian SEM can better reflect substantive theory, can incorporate external prior information, and can in some circumstances outperform maximum likelihood with small samples (Van de Schoot et al., 2015), though MacCallum et al. (2012) and a subsequent systematic review (Van de Schoot et al., 2019) have cautioned that diffuse, non-informative default priors do not automatically solve small-sample problems and can themselves introduce bias unless priors are chosen thoughtfully.

Evaluating Model Fit

Because a structural equation model is a hypothesis about how a set of observed covariances arise, model evaluation centers on comparing the covariance matrix implied by the estimated model with the covariance matrix actually observed in the sample. The chi-square test of exact fit is the traditional statistic for this comparison, but it is well known to be sensitive to sample size, rejecting even trivially misspecified models when samples are large. For this reason, researchers routinely supplement the chi-square test with approximate fit indices.

The Root Mean Square Error of Approximation (RMSEA) estimates the degree of model misspecification per degree of freedom, adjusting the noncentrality parameter $\hat{\lambda} = \max(\chi^2 - df, 0)$ for sample size and model complexity:

$$RMSEA = \sqrt{\frac{\max\left(\frac{\chi^2 - df}{df}, 0\right)}{N - 1}}. \quad (13)$$

The Comparative Fit Index (CFI) and Tucker–Lewis Index (TLI, also called the Non-Normed Fit Index) compare the noncentrality of the hypothesized model, $\hat{\lambda}_M = \chi_M^2 - df_M$, to that of a baseline (null) model of complete independence among the observed variables, $\hat{\lambda}_0 = \chi_0^2 - df_0$:

$$CFI = 1 - \frac{\max(\hat{\lambda}_M, 0)}{\max(\hat{\lambda}_M, \hat{\lambda}_0, 0)}, \quad (14)$$

$$TLI = \frac{\frac{\chi_0^2}{df_0} - \frac{\chi_M^2}{df_M}}{\frac{\chi_0^2}{df_0} - 1}. \quad (15)$$

The Standardized Root Mean Square Residual (SRMR) is based on the discrepancy between observed and model-implied correlations r_{ij} and $\hat{r}_{ij}$ (rather than raw covariances), averaged over all $p(p+1)/2$ unique pairs of the p observed variables:

$$SRMR = \sqrt{\frac{2 \sum_{i=1}^p \sum_{j=1}^i (r_{ij} - \hat{r}_{ij})^2}{p(p+1)}}. \quad (16)$$

The most widely reported of these indices are CFI and TLI, which compare the fit of the specified model to that of the baseline independence model; RMSEA, which penalizes model complexity through df ; and SRMR, which is most directly interpretable as an average standardized residual.

The influential simulation study of Hu and Bentler (1999) evaluated a range of candidate cutoff values for these indices and recommended, as an approximate joint criterion, CFI and TLI values near .95 together with an SRMR value near .08–.09 and an RMSEA value near .06, while explicitly cautioning that these should be regarded as rough guidelines rather than rigid pass/fail thresholds. Later methodological work has both applied and critiqued these benchmarks: Schermelleh-Engel et al. (2003) argued that a two-index reporting strategy combining SRMR with a comparative index such as CFI or RMSEA offers a more balanced assessment than any single index, while Steiger (2007) emphasized that fit index behavior depends heavily on model complexity, sample size, and the type of misspecification present, so that reflexively applying Hu and Bentler's cutoffs to every model can be misleading.

MAJOR VARIANTS AND EXTENSIONS OF SEM

Confirmatory Factor Analysis

Confirmatory factor analysis (CFA) is the measurement-only special case of SEM, used to test whether a hypothesized number of latent factors adequately accounts for the covariances among a set of observed indicators, and to establish the validity and reliability of multi-item scales before those scales are used in a larger structural model (Jöreskog, 1969). CFA is foundational to nearly every other SEM application, since a structural model is only as trustworthy as the measurement model on which it rests.

Path Analysis

Path analysis applies the structural, relational logic of SEM to models composed entirely of observed (manifest) variables, without latent factors (Duncan, 1966; Wright, 1934). It remains useful when constructs are measured with single, well-validated indicators or when the research question concerns the pattern of direct and indirect effects among directly measured variables.

Full (Hybrid) SEM

Full SEM combines the measurement and structural components into a single simultaneously estimated model, following the recommended two-step approach of first validating the measurement model via CFA before estimating structural paths (Anderson and Gerbing, 1988). This is the form most commonly implied when researchers refer simply to “SEM.”

Multi-Group SEM and Measurement Invariance

Multi-group SEM extends the basic model to compare structural relationships, and the measurement properties underlying them, across two or more populations – for example, across countries, gender groups, or experimental conditions. A prerequisite for meaningful group comparisons is measurement invariance, a stepwise sequence of tests establishing that factor loadings, intercepts, and residual variances operate equivalently across groups before structural paths or latent means are compared.

Longitudinal and Growth Models

When the same constructs are measured repeatedly over time, SEM can be extended to latent growth curve models, which represent individual trajectories of change as latent intercept and slope factors, and to cross-lagged panel and autoregressive models, which examine reciprocal influences between variables measured at multiple occasions.

Mediation and Moderation

SEM provides a natural framework for testing mediation, in which the effect of one variable on another is transmitted through one or more intervening variables, by explicitly estimating indirect and total effects and their associated confidence intervals, commonly via bootstrapping procedures. Moderation, in which the strength or direction of a relationship depends on a third variable, can also be incorporated through latent interaction terms or multi-group comparisons.

Composite-Based (PLS) SEM

Partial least squares SEM (PLS-SEM), rooted in Herman Wold's work (Wold, 1982) and popularized in applied business and marketing research through the contributions of Wynne Chin (Chin, 1998) and, later, Joseph Hair and colleagues (Hair et al., 2017, 2011), represents an alternative to the traditional covariance-based SEM (CB-SEM) approach associated with Jöreskog. Rather than estimating latent common factors and reproducing the full observed covariance matrix, PLS-SEM constructs each latent variable η_j as a weighted composite (linear combination) of its own indicators x_{jk} ,

$$\eta_j = \sum_{k=1}^{K_j} w_{jk} x_{jk}, \quad (17)$$

where the outer weights w_{jk} are estimated iteratively so as to maximize the variance explained (R^2) in the endogenous composites of the inner (structural) model,

$$\eta_j = \sum_{i \neq j} \beta_{ji} \eta_i + \zeta_j, \quad (18)$$

rather than by minimizing a discrepancy function such as Equation (10) between $\Sigma(\theta)$ and $\mathbf{S}$. Because it imposes fewer distributional assumptions, tolerates smaller sample sizes, and readily accommodates formatively specified constructs, PLS-SEM has become especially popular in exploratory, prediction-oriented business and marketing research, even as some methodologists continue to debate its suitability as a substitute for CB-SEM in strictly confirmatory, theory-testing contexts (Sarstedt et al., 2016). Recent methodological work has moved away from framing the choice as an either/or decision, instead recommending multimethod strategies that apply both factor-based and composite-based estimators to the same model as a robustness check (Guenther et al., 2025).

Bayesian SEM

As noted above, Bayesian SEM (BSEM) reformulates estimation within a Bayesian framework, producing full posterior distributions for each parameter rather than single point estimates, and allowing researchers to incorporate substantive prior knowledge (Muthén and Asparouhov, 2012). BSEM is particularly attractive for complex models – such as those with many cross-loadings or correlated residuals – that would be under-identified under a strict maximum-likelihood zero-constraint approach, and for applications with small samples, though its properties in that context depend heavily on the informativeness of the chosen priors (Van de Schoot et al., 2019).

Exploratory Structural Equation Modeling and Network Approaches

Exploratory structural equation modeling (ESEM) blends the flexible, all-loadings-estimated character of exploratory factor analysis with the fit evaluation and structural modeling capabilities of confirmatory SEM, offering a middle ground when researchers are not confident that indicators load exclusively on a single factor (Marsh

et al., 2009). In parallel, a growing body of work has explored network psychometric approaches, which model constructs as systems of interacting observed variables rather than as reflections of a single underlying latent factor (Borsboom et al., 2003). Epskamp et al. (2018) proposed a generalized framework unifying network and latent-variable models, prompting ongoing methodological debate about when a latent-variable representation versus a network representation better represents psychological and social phenomena.

SOFTWARE FOR STRUCTURAL EQUATION MODELING

A variety of software packages support SEM, differing in cost, programming interface, and specialized capabilities. Historically dominant commercial packages include LISREL, the original SEM program developed by Jöreskog and Sörbom (Jöreskog and Sörbom, 1978); EQS; IBM SPSS Amos, known for its graphical path-diagram interface; and Mplus, widely used for its extensive support of categorical outcomes, mixture modeling, multi-level SEM, and Bayesian estimation (Muthén and Asparouhov, 2012). SAS offers SEM capabilities through PROC CALIS.

Within the open-source R environment, the `lavaan` package, developed by Yves Rosseel and first released in 2010, has become one of the most widely used tools for SEM, offering a compact model syntax, support for non-normal and missing data, and an option to mimic the numerical output of Mplus or EQS to ease transitions between programs (Rosseel, 2012). Other R-based tools include the `sem` package and OpenMx. For composite-based modeling, SmartPLS is the most widely used dedicated PLS-SEM software (Hair et al., 2017), but the R ecosystem now offers several actively maintained alternatives: `SEMinR`, developed by Ray and Danks, provides a domain-specific syntax that lets users build PLS path models using familiar terms such as `composite()`, `constructs()`, and `paths()`, and additionally allows CB-SEM estimation by delegating to `lavaan` internally (Danks and Ray, 2020; Hair et al., 2021); `cSEM`, developed by Rademaker and Schubert, implements a broad family of composite-based estimators, including consistent PLS (PLSc) and generalized structured component analysis (Rademaker and Schubert, 2020); and `blavaan`, developed by Merkle and Rosseel, extends `lavaan` syntax to Bayesian estimation via JAGS or Stan, giving users access to posterior summaries, Bayesian fit measures, and MCMC diagnostics without leaving the `lavaan` modeling idiom (Merkle et al., 2021; Merkle and Rosseel, 2018). The proliferation of free, open-source tools such as `lavaan`, `SEMinR`, `cSEM`, and `blavaan` has been credited with lowering the barrier to entry for SEM, both for research and for teaching, relative to the earlier era in which state-of-the-art SEM software was largely closed-source and commercial.

APPLICATIONS ACROSS DISCIPLINES

SEM's capacity to model latent constructs and multiple simultaneous relationships has made it a standard analytic tool across a wide range of fields (MacCallum and

Software	License	Notable strengths
LISREL	Commercial	Original SEM program; historical standard (Jöreskog and Sörbom, 1978)
EQS	Commercial	Robust estimators for non-normal data
AMOS	Commercial	Graphical path-diagram interface
Mplus	Commercial	Mixture models, categorical outcomes, Bayesian SEM, multilevel and dynamic SEM (Asparouhov et al., 2018)
lavaan (R)	Open source	Compact syntax; free; mimics Mplus/EQS output (Rosseel, 2012)
blavaan (R)	Open source	Bayesian SEM via JAGS/Stan on top of lavaan syntax (Merkle and Rosseel, 2018)
SEMinR (R)	Open source	Friendly PLS-SEM and CB-SEM syntax (Danks and Ray, 2020)
cSEM (R)	Open source	PLSc and other composite-based estimators (Rademaker and Schubert, 2020)
SmartPLS	Commercial/free tiers	Leading dedicated PLS-SEM GUI software (Hair et al., 2017)

Table 1. Selected software packages for structural equation modeling.

Austin, 2000).

- **Psychology and education:** SEM is used extensively to validate psychological scales through CFA, to test theories of cognitive, motivational, and personality structure, and to model longitudinal change in learning and development through growth curve and cross-lagged designs.
- **Business, management, and marketing:** SEM, and particularly PLS-SEM, is used to test models of consumer behavior, technology acceptance, brand equity, and organizational performance, often linking multiple latent constructs such as satisfaction, trust, and loyalty (Hair et al., 2011).
- **Health and medical research:** SEM supports the validation of patient-reported outcome measures, the modeling of quality-of-life constructs, and the analysis of complex causal pathways linking risk factors, psychosocial mediators, and health outcomes.
- **Sociology, political science, and economics:** consistent with its historical roots, SEM continues to be used to model social stratification, attitudes, and simultaneous-equation systems in which variables mutually influence one another (Blalock, 1964; Duncan, 1966).

STRENGTHS AND LIMITATIONS

The central strength of SEM is its ability to represent theory directly in statistical form: researchers can specify a network of hypothesized relationships, including latent constructs and measurement error, and formally test how well that theory accounts for the observed data. SEM also permits simultaneous estimation of multiple, inter-related equations, decomposition of total effects into direct and indirect components, comparison of competing theoretical models, and testing of measurement equivalence across groups and over time (Anderson and Gerbing, 1988).

These strengths come with important limitations. SEM generally requires relatively large samples for stable maximum-likelihood estimation, particularly as model complexity grows (Jackson, 2003; Kline, 2015), although

Bayesian and composite-based alternatives can relax this requirement under some conditions (Van de Schoot et al., 2015). Model fit statistics assess how well a specified model reproduces observed covariances but cannot, by themselves, establish that the hypothesized causal structure is correct, since statistically equivalent models with different causal interpretations can often fit the same data equally well. The practice of iteratively modifying a poorly fitting model based on sample-specific modification indices, if not cross-validated, risks capitalizing on chance characteristics of a particular sample rather than reflecting a generalizable structure. Finally, the debate over covariance-based versus composite-based (PLS) estimation (Guenther et al., 2025; Sarstedt et al., 2016), and over the appropriate use and interpretation of fit index cutoffs (Hu and Bentler, 1999; Steiger, 2007), illustrates that SEM is not a single settled method but an evolving family of techniques whose proper application requires careful attention to a study's theoretical goals and data characteristics.

RECENT DEVELOPMENTS AND THE EXPANDING SEM ECOSYSTEM

Beyond the broad trends already noted, several concrete methodological and software developments have reshaped SEM practice over roughly the past decade and continue to evolve rapidly.

Dynamic Structural Equation Modeling

Traditional SEM assumes independent observations and is not designed for the densely repeated measurements now common in ecological momentary assessment, experience sampling, and wearable-sensor studies. Asparouhov et al. (2018) introduced dynamic structural equation modeling (DSEM), a general framework that combines multilevel modeling, time-series (vector autoregressive) modeling, and conventional SEM to analyze such *intensive longitudinal data*, where many individuals are each measured repeatedly over many occasions. In a two-level DSEM, within-person dynamics – such as autoregressive and cross-lagged effects among momentary variables – are modeled at Level 1, while individual differences in those dynamic parameters (e.g., a person's average level or their degree of emotional inertia) be-

come random effects predicted by, or predicting, Level-2 constructs. DSEM is typically estimated via Bayesian Markov chain Monte Carlo methods, as implemented in Mplus, and has been applied to model phenomena such as affective dynamics in daily life (Hamaker et al., 2018). This represents one of the more substantial recent extensions of the SEM framework, bringing it into closer contact with time-series econometrics and multilevel modeling.

Integration with Machine Learning

A growing body of work explores hybrid SEM-machine learning approaches that combine SEM's capacity for theory-driven, interpretable modeling of latent constructs with machine learning's flexibility in capturing nonlinear patterns and improving out-of-sample predictive accuracy. Richter and Tudoran (2024) discuss how PLS-SEM results can be complemented with machine learning algorithms (e.g., random forests or gradient boosting) to detect nonlinearities, interaction effects, and important predictors that a linear structural model might miss, while still retaining SEM's structured, construct-based interpretation. Other applications use classification algorithms alongside CFA to jointly optimize measurement-scale fit and predictive accuracy when shortening psychometric scales. At a more structural level, packages such as SEMdeep and SEMgraph implement SEM-constrained neural networks and causal network analysis, in which the path structure specified by a substantive SEM (or a graph learned from high-throughput data) is used to constrain the connectivity of a deep neural network, combining the interpretability of a specified causal structure with the flexible function approximation of deep learning (Grassi et al., 2022). These hybrid approaches remain an active and still-maturing area, and questions about how to evaluate fit, interpret parameters, and avoid overfitting in such combined models are the subject of ongoing methodological work.

Expanding the R Software Ecosystem

As detailed in Section 5, the R ecosystem for SEM has expanded substantially in recent years beyond lavaan alone. SEMinR (Danks and Ray, 2020; Hair et al., 2021) and cSEM (Rademaker and Schuberth, 2020) have made composite-based (PLS) modeling more accessible in R, an environment previously dominated by CB-SEM tools, and both packages implement newer composite-based estimators such as consistent PLS (PLSc), designed to correct for the attenuation bias that ordinary PLS weights can introduce when constructs are meant to represent common factors. blavaan (Merkle et al., 2021; Merkle and Rosseel, 2018) has similarly extended Bayesian SEM into the open-source R environment using familiar lavaan syntax, reducing reliance on proprietary Bayesian SEM implementations. Companion packages such as semTools provide auxiliary functions for tasks including measurement invariance testing, reliability estimation, and power analysis, further consolidating the open-source SEM toolchain.

Multimethod and Robustness-Oriented Practice

As noted in Section 4.7, recent methodological guidance increasingly frames CB-SEM and PLS-SEM as complementary rather than competing tools, encouraging researchers to report results from both factor-based and composite-based estimators on the same model as a robustness check rather than committing exclusively to one estimation tradition (Guenther et al., 2025). A related development applies double-machine-learning-style robustness checks – originally developed in the econometric causal-inference literature – to re-examine whether theory-specified SEM paths remain directionally and statistically stable when flexible machine-learning estimators are used to adjust for controls and covariates, offering an additional diagnostic layer beyond traditional fit indices.

CURRENT TRENDS AND FUTURE DIRECTIONS

Several broader trends are shaping the future trajectory of SEM research, building on the specific developments above. First, the multimethod integration of CB-SEM and PLS-SEM, and increasingly of SEM and machine learning more broadly, is likely to continue as computational tools make it easier to fit and compare multiple modeling traditions on the same data. Second, Bayesian methods, including Bayesian regularization, small-variance priors for approximate measurement invariance, and dynamic SEM's Bayesian estimation of intensive longitudinal data, continue to be refined, alongside ongoing methodological scrutiny of how sensitive results are to prior specification (MacCallum et al., 2012; Van de Schoot et al., 2019). Third, exploratory structural equation modeling and related flexible measurement approaches are being used increasingly to relax rigid simple-structure assumptions in confirmatory models (Marsh et al., 2009). Fourth, improved tools for a priori power analysis targeted at specific parameters, rather than global model fit alone, are making principled sample-size planning more accessible to applied researchers (Wang and Rhemtulla, 2021). Finally, the relationship between traditional latent-variable SEM and network psychometric approaches remains an active area of theoretical and methodological debate (Borsboom et al., 2003; Epskamp et al., 2018), with implications for how psychological and social constructs are conceptualized and measured going forward.

CONCLUSION

Structural equation modeling has evolved over more than a century, from Sewall Wright's biometric path diagrams and the psychometric factor-analytic tradition of Spearman and Thurstone, through Karl Jöreskog's unification of measurement and structural modeling and the diffusion enabled by the LISREL program, into today's diverse ecosystem of covariance-based, composite-based, dynamic, and Bayesian approaches implemented across both commercial software and an increasingly capable open-source R toolchain (lavaan, blavaan, SEMinR, cSEM). Its capacity to formally link theoretical constructs

to empirical indicators while modeling measurement error and complex, multi-equation relationships explains its enduring and expanding role across the social, behavioral, health, and management sciences. Recent extensions – dynamic SEM for intensive longitudinal data, hybrid SEM–machine learning frameworks, and multi-method CB-SEM/PLS-SEM practice – show that the technique continues to absorb and adapt tools from adjacent fields rather than standing still. At the same time, ongoing debates over estimation approach, fit evaluation, sample size and power, and the interpretation of latent variables underscore that SEM remains an active and evolving methodological field rather than a fixed, uncontested technique. Researchers using SEM are therefore best served by grounding their modeling choices in both their substantive theory and a clear understanding of the assumptions and trade-offs associated with the particular SEM variant, estimator, and software tool they adopt.

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